

Santhosh Kothapalli

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ABOUT ME

Systems Engineer specializing in robotics and automation, with hands-on experience in collaborative robots, control systems, and embedded programming.

Skills

- **Programming & Middleware:** C++, Python, ROS2
- **Controls & Autonomy:** PID, MPC, State Machines, Behavior Trees, Visual Servoing, Reinforcement Learning
- **Simulation & Design:** MATLAB, Gazebo, MuJoCo | SolidWorks, CREO, Autodesk Fusion
- **Embedded & Electronics:** Sensor Fusion, EtherCAT, UART, I2C, SPI

PROFESSIONAL EXPERIENCE

Robotics Engineer | Svaya Robotics July 2023 – Present | Hyderabad, India

Six and Seven Axis Robotic Arms

- Engineered and optimized custom Joint Torque Sensors (JTS) and 6-axis F/T sensors across multiple load capacities, balancing structural compliance and weight to achieve sub-1% accuracy.
- Developed and deployed core force-control algorithms (Impedance, Free-Drive, and Hybrid Force-Position control), introducing an adaptive assembly feature that significantly reduced robot downtime and operator intervention in high-precision tasks on production lines.
- Modeled compliant actuators to jointly optimize structural compliance and control-loop response, reducing position settling time by 50% for high-speed, high-precision industrial applications.
- Implemented real-time collision detection and safety response systems, designing a custom impact-force measurement device to ensure strict compliance with ISO cobot safety benchmarking.
- Architected automated calibration sequences, test rigs, and validation workflows, scaling assembly/inspection success rates to >90% while bridging field-deployment insights back to the R&D pipeline.

Mechanical Engineering Intern, Mowito Automation

May 2022 – July 2022 | Bengaluru

Automated storage – retrieval systems for warehouses

- Simulated and optimized multi-robot ASRS operations using FlexSim to evaluate warehouse throughput and identify kinematic bottlenecks.
- Translated simulation insights into hardware requirements, designing belt-driven linear actuators tailored to specific velocity and acceleration constraints.
- Designed robust sliding contact power distribution systems to support long-endurance autonomous warehouse operations.

PROJECTS

Adaptive Robotic Assembly System (Professional & Personal Research)

- Developed a virtual Force-Torque Sensor using joint torque sensor data on a 6-DOF manipulator to enable compliant assembly behaviours.
- Implemented an Image-Based Visual Servoing (IBVS) pipeline in Python/C++, integrating real-time camera tracking with force feedback.
- Designed a Sensor Fusion architecture to combine high-frequency force/torque data with vision matrices, achieving precision alignment and assembly task completion.

Autonomous Self-Balancing Mobile Robot (Classical control, RL and hardware)

- Modelled a two 8-DOF legged wheeled robot in MuJoCo sim and developed a control architecture to achieve core functionality including translational velocity, yaw control and height adjustment thorough squatting
- Designed and fabricated a two-wheeled inverted pendulum robot, deriving the system dynamics and plant model via Frequency Response Analysis.
- Implemented a classical PID controller for real-time stabilization based on the empirical plant model.
- Deployed a Reinforcement Learning (RL) policy directly onto the physical hardware, mapping IMU state inputs to motor voltage outputs to benchmark modern vs. classical control performance.

BEEBOT: Winged Micro Aerial Vehicle (MAV) (Simulation & Bio-Inspired Design)

- Designed a high-frequency compliant transmission mechanism to convert linear piezoelectric actuator oscillations into bio-mimetic wing-flapping kinematics.
- Modeled structural fatigue and resonance frequencies using FEA tools to optimize power transmission efficiency.
- Developed a PID control loop to regulate aerodynamic thrust, pitch, and roll based on sensor feedback.

Double Wishbone Suspension Optimization (Simulation & Math)

- Designed a full-car double wishbone suspension system and modelled its multi-body dynamics in **MATLAB**.
- Applied a Genetic Algorithm (GA) to iteratively optimize suspension geometry, achieving a massive reduction in vehicle roll and bounce over simulated rough terrain.

Education and Leadership

National Institute of Technology, Warangal

Telangana, India

Bachelor of Technology in Mechanical Engineering

CGPA – 7.50

2019- 2023

Hardware Team Captain, ROBOCON 2022: Led a multi-disciplinary team to design and manufacture two autonomous/semi-autonomous ring-throwing robots; achieved a 96/100 score in Stage-1 qualifiers.